## Experiment 16:

## DFT, Circular Convolution and Linear/Sectional Convolution

### OBJECTIVES

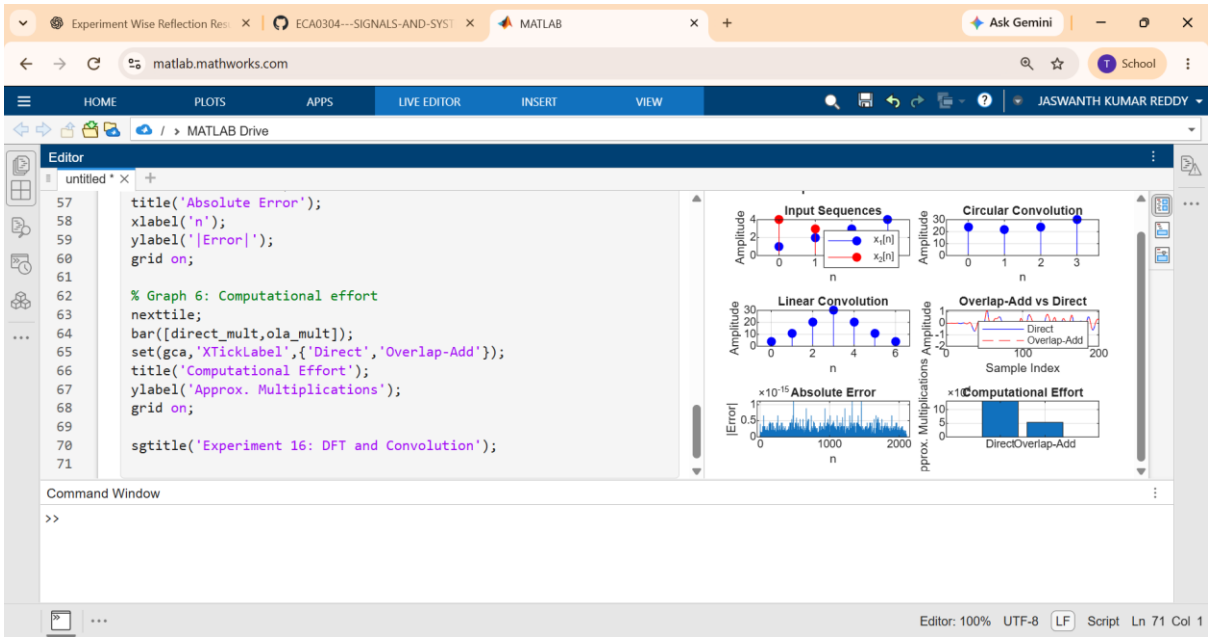
• Compute the N-point DFT of a sequence and verify Parseval's relation.

• Compute circular convolution directly and through DFT multiplication.

• Compare circular convolution with linear convolution and determine the required zero-padding length.

• Implement sectional convolution using the Overlap-Add method for long sequences.

### Output Graphs:




**Result:** The discrete Fourier transform (DFT), circular convolution, and linear or sectional convolution operations were performed and analyzed using MATLAB.